

Omin Kwon

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RESEARCH INTEREST

Computer Architectures & Machine Learning Systems, with an emphasis on co-designing ML algorithms and computer systems for efficient ML.

EDUCATION

Seoul National University

Sept 2025 – present

M.S. in Computer Science and Engineering

- Advisor: Prof. Jae W. Lee, ARC Lab
- GPA: 4.3/4.3

Seoul National University

Mar 2019 – Aug 2025

B.S. in Electrical and Computer Engineering (summa cum laude)

- GPA: 4.26/4.3 (upper-division), 3.95/4.3 (overall)
- Leave of absence for mandatory military service (May 2021-Feb 2023)

PUBLICATIONS

HERALD: High Throughput Block Diffusion LLM Serving via CPU-GPU Cooperative KV Cache Retrieval

Omin Kwon, Doyeon Kim, Jongseok Park, Seung Yul Lee, Ion Stoica, Jae W. Lee

Under review

Paper Link: arxiv.org/abs/2606.21633

ELIXIR: Disaggregating FFN, Attention, and KV Cache via CXL-PNM for Elastic LLM Serving

Omin Kwon*, Kyungsoo Kim*, Ugyeong Song, Yeonhong Park, Nachuan Wang, Namsung Kim, Jae W. Lee

Under review

MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM

Omin Kwon, Yeonjae Kim, Doyeon Kim, Minseo Kim, Yeonhong Park, Jae W. Lee

ICML 2026 AdaptFM Workshop (Oral); extended version under review

Paper Link: arxiv.org/abs/2602.14209

STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer

Omin Kwon*, Kyungjun Oh*, Jaeyong Lee, Myungsuk Kim, Jihong Kim

ICCAD 2025

Paper Link: doi.org/10.1109/ICCAD66269.2025.11240772

NestedFP: High-Performance, Memory-Efficient Dual-Precision Floating Point Support for LLMs

Haeun Lee, Omin Kwon, Yeonhong Park, Jae W. Lee

NeurIPS 2025

Paper Link: neurips.cc/virtual/2025/loc/san-diego/poster/117613

AiDE: Attention-FFN Disaggregated Execution for Cost-Effective LLM Decoding on CXL-PNM

KyungSoo Kim, Omin Kwon, Yeonhong Park, Jae W. Lee

IEEE Computer Architecture Letters (CAL)

Paper Link: doi.org/10.1109/LCA.2025.3597323

EXPERIENCES

Visiting Researcher

Mar 2026 – May 2026

University of Illinois Urbana-Champaign (UIUC)

Urbana, IL, USA

- FAST Lab, hosted by Prof. Nam Sung Kim

Microelectronics (430.207B) Tutor Seoul National University	Mar 2025 – June 2025 Seoul, South Korea
Digital System Design (430.315) Tutor Seoul National University	Sept 2024 – Dec 2024 Seoul, South Korea
Computer Organization and Design (430.322) Tutor Seoul National University	Mar 2024 – June 2024 Seoul, South Korea
Military Service Republic of Korea Air Force	May 2021 – Feb 2023 South Korea

PRESENTATIONS

MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM Oral paper at the ICML 2026 AdaptFM Workshop (1.5%). arXiv.	2026
STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer Oral presentation at ICCAD 2025, Munich. Paper.	2025

HONORS & AWARDS

Grand Prize (Minister of Science and ICT Award), 2024 AI Semiconductor Idea and Design Contest Ministry of Science and ICT Designed a resolution-aware diffusion accelerator leveraging computational staleness. News: Edaily.	Jan 2025
Grand Prize (Deputy Prime Minister and Minister of Education Award), 2024 STOB League Semiconductor Contest Ministry of Education Designed a state-aware data randomizer for high-performance SSDs, improving performance and device lifetime. News: Dong-A Ilbo.	Dec 2024
Grand Prize, 2024 FriendlyAI LLM Hackathon FriendlyAI Developed a knowledge-graph RAG application using the FriendlyAI API.	July 2024
SNU Specialized Semiconductor College Scholarship Seoul National University	Aug 2024 – Aug 2025
Merit-Based Scholarship Seoul National University	2023, 2024